

1 **Spatio-temporal variability in slab temperature within**
2 **dynamic 3-D subduction models**

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Summary

Spatio-temporal variability in arc geochemistry and the conditions recorded by exhumed rocks suggest subduction zone thermal structure evolves in time and along-strike. Although much effort has been dedicated to studying subduction zone thermal structure, we lack an understanding of spatio-temporal temperature variability during time-dependent subduction. We model three-dimensional, dynamic subduction and examine the time evolution of the along-strike temperature difference of the slab’s upper surface (“slab-top”) at the center relative to the edge of the subduction zone. We examine this slab-top temperature variability for subduction systems of different widths and with different plate mobilities (i.e., fixed vs. free subducting and overriding plates). In all of our models, the main control on slab-top temperature is convergence rate; either by simply controlling the rate of slab sinking or via the effect it has on the decoupling depth (DD). In the early stages of subduction, more rapid convergence at the plate center produces a cooler slab relative to warmer slab edges. For mature subduction, this flips; a shallower DD at the slab center produces warmer temperatures with respect to the edge. Importantly, our maximum along-strike temperature changes are reduced ($\leq 50^{\circ}\text{C}$) relative to previous kinematically-driven modeling studies, due to a reduced role for slab-top heating via toroidal flow. Our dynamic subduction models, therefore, point towards a strong time dependence in the sense of along-strike temperature variation, but with relatively low absolute values in geometrically simple subduction zones.

Keywords

Subduction zone processes; Continental margins: convergent; Numerical modelling; Dynamics of lithosphere and mantle

1 Introduction

Subduction zone thermal structure regulates a plethora of important geological processes, including arc volcanism (e.g., Katili, 1975) and the metamorphism of downgoing rocks (e.g., Ernst, 1973). At many subduction zones, these geologic products suggest significant spatial and temporal variability in subduction zone temperature fields: The pressure-temperature (P-T) conditions of exhumed metamorphic rocks can record significant variation in thermal history between units that are spatially close (e.g., Platt, 1975; O’Brien, 1997), and exhumed rocks associated with oceanic terrains suggests highly

variable P-T conditions during different stages of subduction, with typically warmer conditions during the earliest phases of subduction that later transition to colder conditions (e.g., Platt, 1975; Cloos, 1985; Krebs et al., 2011; Agard et al., 2018). The production rate and geochemistry of arc magma has also been shown to be time-dependent (e.g., Tupinambá et al., 2012; Elburg & Foden, 1998) and exhibit along-arc variability which, at numerous subduction zones, has been attributed to thermal variability of the slab and/or mantle wedge (e.g., Portnyagin & Manea, 2008; Bai et al., 2020).

Such observations suggest that the thermal conditions of slabs are both temporally and spatially variable. Numerical subduction models are often used to investigate the mechanisms behind slab temperature variability. Previous numerical models have shown that basic subduction zone properties – convergence rate, slab dip, and downgoing plate age – exert a first-order control on temperatures along the surface of the slab or the “slab-top” (e.g., Peacock, 1996; Peacock & Wang, 1999; Van Keken et al., 2002). To this end, the product of these three properties, termed the “thermal parameter” (Kirby et al., 1996), has been utilized as a first-order indicator of slab-top temperatures with low and high values corresponding to warm and cold subduction zones respectively (e.g., van Keken et al., 2011).

In this modeling study, we utilize and extend these fundamental principles to probe the thermal evolution of 3-D subduction zones. Previous numerical subduction models, which specifically target subduction zone thermal structure, are either dynamic but 2-D (e.g., Kincaid & Sacks, 1997; Holt & Condit, 2021; Zhou & Wada, 2021) or are 3-D but kinematically impose the behavior of part of the subduction system (e.g., Bengtson & van Keken, 2012; Morishige & van Keken, 2014; Wada et al., 2015; Plunder et al., 2018). Kinematically-driven thermal models, in which quantities such as convergence rate and slab geometry are prescribed at all times, are well suited to describe specific Earth subduction zones with known kinematic properties. When constructed in 3-D, such models have illuminated a range of subduction processes that can produce along-strike temperature variability within time-invariant subduction zones: E.g., mantle flow regime and vigor (Wada, 2021; Wada et al., 2015; Bengtson & van Keken, 2012; Kincaid & Griffiths, 2004), along-strike variability in slab-rollback rate and convergence rate (Ji et al., 2017; Bengtson & van Keken, 2012; Plunder et al., 2018), trench obliquity and decoupling depth (Morishige & van Keken, 2014), and complex plate and slab geometries (Ji et al., 2016). These studies highlight a wide range of along-strike slab-top temperature changes, span-

ning values from 30°C (Plunder et al., 2018) to 200°C (e.g., Bengtson & van Keken, 2012; Wada et al., 2015; Ji et al., 2016; Plunder et al., 2018), with intermediate values of $\sim 100^\circ\text{C}$ (e.g., Morishige & van Keken, 2014, Kincaid & Griffiths, 2004) as a result of the above-mentioned processes. These studies, however, do not account for temporal variability.

Modeling the temporal evolution of subduction zone thermal structure has been mainly limited to 2-D (e.g., Kincaid & Sacks, 1997; Holt & Condit, 2021; Zhou & Wada, 2021). By enabling the subduction system to evolve self-consistently in “dynamic” models, Holt & Condit (2021) demonstrated that the slab-top cools during the early and intermediate phases of subduction, in basic agreement with peak P-T conditions extracted from the exhumed rock record. The primary drivers of this thermal evolution are temporal changes in convergence rate and decoupling depth (DD). The DD marks the down-dip transition from cold and stiff material directly above the slab (i.e., from the forearc region of the upper plate) to hot and warm material above the slab (i.e., from the circulating portion of the mantle wedge). From a mechanical perspective, this corresponds to a transition from a slab that is decoupled from overlying material, and so does not drag the forearc down the subduction zone, to a slab that is coupled to the weak mantle wedge and hence exerts shear stresses that drive wedge flow (Furukawa, 1993). In kinematically-driven thermal models, it is typically held fixed at ≈ 80 km depth (Wada & Wang, 2009; Syracuse et al., 2010). Dynamic models of the type considered here, and within which the DD evolves self-consistently (albeit up to some maximum, imposed depth), therefore enable this additional mechanism to drive time-dependent slab thermal structure.

In this work, we use a 3-D and dynamic subduction modeling approach to explore how slab-top temperature (i.e., the temperature extracted at the top of the oceanic crust comprising the uppermost layer of the slab) varies in the trench-parallel direction, and how this variation evolves over tens of millions of years. This approach allows us to simultaneously account for the temporal and spatial variability suggested by natural observations and, in turn, probe the physical mechanisms responsible for such variability in our models. We model dynamic subduction using the ASPECT finite element code (Kronbichler et al., 2012; Heister et al., 2017; Bangerth et al., 2019). We examine the effect of plate/slab width (1000 km, 2000 km, infinite) and plate mobility (“fixed” vs. “free” subducting and overriding plates) on the dynamics of subduction and how these variations manifest in spatio-temporal slab-top temperature changes. We extract slab-top temperatures for three different stages (free sinking, interaction with the lower man-

tle, mature subduction) and depths (60, 130, 200 km), and focus on the role of convergence rate, mantle flow pattern and DD variability in driving slab temperature variability. Finally, we place our results into the context of previous modeling and observational studies that target subduction zone thermal structure.

2 Methods

2.1 Governing Equations

We model subduction in a three-dimensional Cartesian domain using the finite-element code ASPECT, version 2.1.0 (Kronbichler et al., 2012; Heister et al., 2017; Bangerth et al., 2019). We solve the conservation equations for mass (equation 1), momentum (equation 2), and energy (equation 3) for a viscous, incompressible Boussinesq fluid with no inertia, shear heating, or internal heating:

$$\nabla \cdot \mathbf{v} = 0 \quad (1)$$

$$-\nabla \cdot 2\eta\dot{\epsilon} + \nabla p = \rho\mathbf{g} \quad (2)$$

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \right) - k \nabla^2 T = 0 \quad (3)$$

where $\mathbf{v}$ is the velocity, η the viscosity, $\dot{\epsilon}$ the strain rate tensor, p the pressure, ρ the density, $\mathbf{g}$ the gravitational acceleration vector, C_p the specific heat capacity, T the temperature and k the thermal conductivity. Specific values for the parameters defined in this section can be found in Table 1.

Our models include three compositional fields that are rheologically distinct from the background material (but have equivalent density). We track and conserve these compositional fields (Equation 4) using the fields method (Bangerth et al., 2021).

$$\frac{\partial C_i}{\partial t} + \mathbf{v} \cdot \nabla C_i = 0 \quad (4)$$

Here, i identifies each individual composition ($i = 1, 2, 3$).

Table 1: List of parameters with values and units.

Parameter	Symbol	Value	Unit
Reference values			
Reference viscosity	η_{ref}	2.5×10^{20}	$\text{Pa} \cdot \text{s}$
Reference strain rate	$\dot{\epsilon}$	5×10^{-15}	s^{-1}
Reference density	ρ_0	3300	kg m^{-3}
Gravitational acceleration	$\mathbf{g}$	9.81	m s^{-2}
Specific heat capacity	C_p	940	J K^{-1}
Surface temperature	T_s	0	$^{\circ}\text{C}$
Mantle potential temperature	T_m	1421	$^{\circ}\text{C}$
Adiabatic gradient		0.3	$^{\circ}/\text{km}$
Thermal conductivity	k	3	$\text{W m}^{-1} \text{K}^{-1}$
Thermal diffusivity	κ	10^{-6}	$\text{m}^2 \text{s}^{-1}$
Thermal expansivity	α	3×10^{-5}	K^{-1}
Rheology: Diffusion and dislocation creep			
Prefactors	A_{diff}	upper mantle diffusion: 1.92×10^{-11}	$\text{Pa}^{-1} \text{s}^{-1}$
		lower mantle diffusion: 1.67×10^{-13}	$\text{Pa}^{-1} \text{s}^{-1}$
	A_{disl}	upper mantle dislocation: 5.858×10^{-18}	$\text{Pa}^{-n} \text{s}^{-1}$
Exponent	n	dislocation: 3.5	-
Activation volumes	$V_{a,diff}$	upper mantle diffusion: 4×10^{-6}	m^3
		lower mantle diffusion: 2.5×10^{-6}	$\text{m}^3 \text{mol}^{-1}$
	$V_{a,disl}$	upper mantle dislocation: 12×10^{-6}	$\text{m}^3 \text{mol}^{-1}$
Activation energies	$E_{a,diff}$	diffusion: 300×10^3	J
	$E_{a,disl}$	dislocation: 540×10^3	J
Gas constant	R	8.3145	$\text{J mol}^{-1} \text{K}^{-1}$
Rheology: Pseudo-plastic yielding			
Friction coefficient	a	0.6	-
Cohesion	b	60	MPa
Prefactor	λ	0.1	-

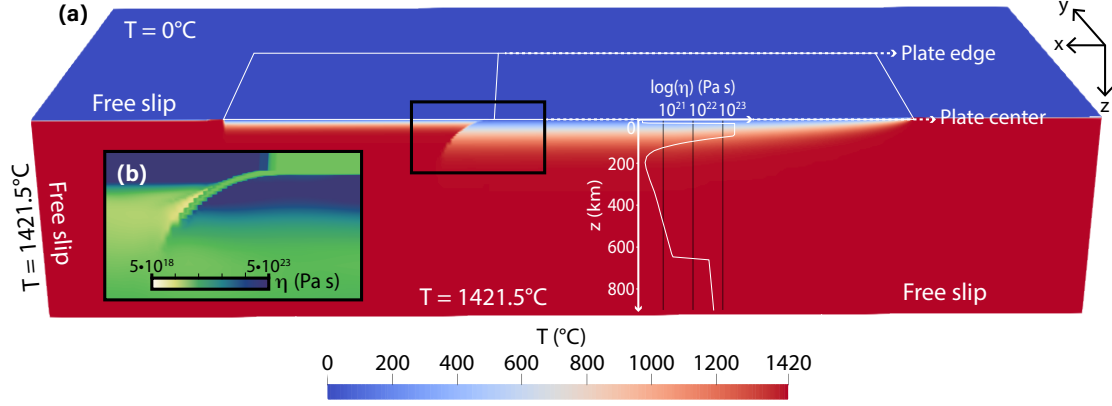


Figure 1: (a) Initial thermal structure for the reference model (with free plates and a 2000 km plate width). (b) The initial viscosity field within the plate interface area. The weak material on top of the subducting plate corresponds to the crustal layer. In panel (a), we also include a depth profile of the model viscosity.

2.2 Model Setup

We consider dynamic intra-oceanic subduction: both the overriding and subducting plates are oceanic, and our system evolves self-consistently for 50 Myr. We vary the width of the plates between 2000 km and 1000 km, and trench mobility is taken into account by fixing either the upper or the lower plate, or by leaving them both free to move. The model domain is a box $(x, y, z) = (4500, 1800, 900)$ km, where x is the length, y the width and z the depth (Figure 1). We take advantage of plate symmetry about the center of the subduction system (i.e., the trench-perpendicular vertical axis at $y = 0$), which enables us to run models with half of the stated plate widths. Our reference model corresponds to a case in which both plates are free to move and their full plate width is 2000 km. Our models include the first 240 km of the stronger lower mantle (cf. Stegman et al., 2006), to better simulate the interaction between the slab and the 660 km depth discontinuity. The initial setup does not include lateral plates, but relatively thin plates form during the model run due to surface cooling and the formation of a stiff thermal boundary layer. This counterbalances in part the significant along-strike deformation of

the active plates that would occur if lateral plates were completely omitted (Yamato et al., 2009). All model boundaries are mechanically free slip.

The temperature is set to $T_s = 0^\circ\text{C}$ at the surface, and the other boundaries are fixed at the mantle potential temperature $T_m = 1421.5^\circ\text{C}$ (GDH1 plate cooling model: Stein & Stein, 1992). The initial thermal structure of the oceanic plates are half-space cooling profiles (Turcotte & Schubert, 2002) for subducting plate and overriding plate ages of 90 Myr and 10 Myr, respectively (see Table 1 for thermal parameter values). Our subduction models are incompressible but we add a $0.3^\circ\text{C}/\text{km}$ adiabatic temperature gradient to the extracted slab top temperatures (cf. van Keken et al., 2011). A weak ($2 \times 10^{20} \text{ Pa} \cdot \text{s}$) and thin (10 km thick) crust, with the same density as the mantle, is embedded into the top of the subducting plate to decouple the upper and lower plates (e.g. Běhouňková & Čížková, 2008; Capitanio et al., 2010; Quinquis et al., 2011). Subduction is initiated by extending the weak crust and subducting plate thermal anomaly to a depth of $\approx 120 \text{ km}$ and with a radius of curvature of 250 km. When juxtaposed next to the thinner (and hence lighter) overriding plate, this provides the initial density anomaly needed to drive subduction. Model densities are exclusively temperature dependent and computed relative to the reference density, $\rho_0 = 3300 \text{ kgm}^{-3}$, of material at the mantle potential temperature: $\rho = \rho_0(1 - \alpha(T - T_m))$.

In addition to the background material, the model subduction zone contains three additional compositions: the crust, a rheologically stronger material that makes up both the overriding plate and a 15 km thick core in the middle of the subducting plate, and weak boxes at the end of each of the free plates. All three are characterized by different rheologies or viscosities (see Section 2.3), but have equivalent densities to the background material. The weak boxes, which are 50 km wide and 50 km deep, assist with maintaining ridge-like thermal structures at the end of each of the free plates. These three components are implemented as different compositional fields, and tracked by solving Equation 4.

An important property of our model is the depth of decoupling (or decoupling depth, DD, determined by computing the depth at which a viscosity contour for $\eta = 10^{21.5} \text{ Pa} \cdot \text{s}$ intersects the slab top), which marks the down-dip beginning of viscous coupling between the slab-top and the hot and circulating mantle wedge (Furukawa, 1993). At this depth, the slab transitions from being overlain by cold (and stiff) forearc material to being over-

lain by warm (and weak) mantle material; this depth is, therefore, also associated with a relatively abrupt down-dip increase in slab-top temperature. At current Earth subduction zones, which mainly correspond to mature phases of subduction, the DD has been estimated by constraining kinematically-driven thermal models with surface heat flow measurements and, following this, hypothesized to be at 60–80 km depth (Furukawa, 1993; Wada & Wang, 2009). In our models, we allow the DD to vary with time as a function of evolving subduction zone properties, but limit it at an imposed maximum decoupling depth (MDD) of 100 km (where we convert the weak oceanic crust to an equivalent strength as the surrounding mantle and hence prohibit slab-mantle coupling beyond this depth). In reality, this MDD is likely associated with a down-dip transition from weak hydrous minerals to stronger anhydrous minerals (e.g., Wada et al., 2008; Hirauchi & Katayama, 2013; Agard et al., 2020; Peacock & Wang, 2021); here, we simplify this by pinning it to a constant depth. The initial DD, corresponding to the depth at which the base of the overriding plate is in contact with the slab-top, is ≈ 30 km: this corresponds to the initial thermal thickness of the 10 Myr old overriding plate. As the model evolves, the DD gradually deepens in a self-consistent manner, in some cases all the way to the imposed MDD limit. This DD increase, from subduction initiation to maturity, is in agreement with inferences from previous modeling and observational studies (e.g., Agard et al., 2020; Holt & Condit, 2021; Wang et al., 2023).

2.3 Rheology

The mantle has a composite diffusion-dislocation creep rheology. The diffusion and dislocation creep flow laws are given by Equations 5 and 6:

$$\eta_{diff} = \frac{1}{A_{diff}} \exp\left(\frac{E_{diff} + pV_{diff}}{RT}\right) \quad (5)$$

$$\eta_{disl} = A_{disl}^{-\frac{1}{n}} \dot{\epsilon}_{II}^{\frac{1-n}{n}} \exp\left(\frac{E_{disl} + pV_{disl}}{nRT}\right) \quad (6)$$

where $A_{diff/disl}$ are the pre-factors for diffusion and dislocation creep, n the dislocation creep power-law exponent, $\dot{\epsilon}_{II}$ is the second invariant of the strain rate, $E_{diff/disl}$ the activation energy, $V_{diff/disl}$ the activation volume, R the gas constant and T the temperature, which includes the prescribed adiabatic gradient of $0.3^\circ/\text{km}$. The values for

these parameters, reported in Table 1, are in accordance with experimental values for dry olivine (Hirth & Kohlstedt, 2003). We compute the prefactors $A_{diff/disl}$ for upper mantle diffusion and dislocation creep such that, at a depth of 330 km and reference strain rate of 10^{-15} s^{-1} , each component has a reference value $\eta_{diff} = \eta_{disl} = 4 \times 10^{20} \text{ Pa} \cdot \text{s}$. When averaged together, this produces a reference upper mantle viscosity of $2 \times 10^{20} \text{ Pa} \cdot \text{s}$. The lower mantle is more viscous and is set to only deform via diffusion creep. The upper to lower mantle viscosity jump is parameterized with a diffusion creep viscosity that is equal to 15 times that of diffusion creep in the lowermost upper mantle. Due to the operation of dislocation creep in the upper mantle, the effective viscosity jump is actually ≈ 30 , as is compatible with geoid constraints (e.g., Hager, 1984).

Throughout the domain, the effective viscosity is then computed by combining these viscosity components with η_{yield} via a harmonic mean:

$$\eta = \left(\frac{1}{\eta_{diff}} + \frac{1}{\eta_{disl}} + \frac{1}{\eta_{yield}} \right)^{-1}. \quad (7)$$

η_{yield} is a plastic ‘viscosity’, which is incorporated to loosely mimic the large-scale weakening effects of brittle faulting at lithospheric depths. This plastic deformation only operates within the lithosphere; at high stresses, η_{yield} reduces to cap the model stress at the prescribed yield stress, τ_{yield} . η_{yield} is therefore calculated as:

$$\eta_{yield} = \frac{\min(\tau_{yield}, 0.5 \text{ GPa})}{2\dot{\epsilon}} \quad (8)$$

where the yield stress τ_{yield} is approximated by a Coulomb friction criterion for optimally oriented faults:

$$\tau_{yield} = (a\sigma_n + b) \lambda \quad (9)$$

where $a = 0.6$ is the friction coefficient, $b = 60 \text{ MPa}$ is the cohesion, σ_n is the lithostatic pressure and $\lambda = 0.1$ is a constant slab weakening prefactor. These values are broadly consistent with previous dynamic subduction modeling studies (e.g., Enns et al., 2005; Di Giuseppe et al., 2008; Holt et al., 2015; Behr et al., 2022). The weak boxes that we add at the edges of the free plates (to decouple them from the adjacent, stiff material of the growing thermal boundary layer) are implemented by reducing the yield stress

pre-factor ($\lambda = 0.05$). The overriding plate and the core in the subducting lithosphere are made strong by switching off yielding in these regions. The viscosity is limited between minimum and maximum values of $2.5 \times 10^{18} \text{ Pa} \cdot \text{s}$ and $2.5 \times 10^{23} \text{ Pa} \cdot \text{s}$.

2.4 Numerical Methods

The ASPECT finite element code includes adaptive mesh refinement (AMR) functionality. In our models, we refine the mesh as a function of viscosity, temperature, and composition: greater refinement occurs in areas where the gradients of these properties are large. This AMR setup produces highly refined areas along the evolving plate interface and within the slab, and a coarser mesh in regions without high viscosity, temperature, or compositional gradients (e.g., the lower mantle). In these latter regions, cubic finite elements are 56 km wide/thick, while in highly refined regions, we reach a maximum resolution of 1.8 km. In addition to this AMR, we fix the upper 100 km of the model domain, and the upper-lower mantle boundary region, to have a minimum resolution of 7 km.

To verify our solver tolerances are sufficiently strict and our model resolution is sufficiently high, we performed numerical accuracy and resolution tests on a 2-D version of our reference model. Relative to the reference setup, we reduced (i) the linear (from 10^{-3} to 10^{-4}) and non-linear (from 5×10^{-3} to 5×10^{-4}) solver tolerances, and (ii) doubled the mesh resolution. We test these changes by comparing the convergence rate evolution. Adjusting the solver tolerances has no discernible impact (Supplementary Figure S1a). Doubling the resolution does not alter the overall trend, but results in minor convergence rate differences that are $< 5\%$ for most of the run (Supplementary Figure S1b).

3 Results

We focus on key aspects of subduction dynamics and their effect on the spatio-temporal evolution of slab-top temperature. In particular, the mantle wedge flow regime, convergence rate, and decoupling depth (DD), all of which significantly impact slab temperatures. In order to generate wide variability in such properties, we examine models with variable plate mobilities (both plates free, fixed subducting plate, fixed overriding plate) and variable plate/slab widths (1000 km width, 2000 km width, infinite width/2-D mod-

els). Fixing the subducting or overriding plate results in, respectively, enhanced or reduced slab rollback (e.g., Enns et al., 2005; Capitanio et al., 2010; Holt et al., 2015). Slab rollback is also affected by plate width, with narrow subduction zones previously associated with faster rollback, more toroidal flow, and higher along-strike gradients in subduction properties (Stegman et al., 2006; Piromallo et al., 2006)

3.1 Reference case

Our reference model contains 2000 km wide plates and free subducting and overriding plates. The temporal evolution of the slab shape and slab-top temperature is shown in Figure 2a.

We identify three main stages of subduction by focusing on time periods with distinct convergence rates (Figure 3a,b). In our analysis, we do not consider thermal structure during the slow subduction initiation phase that occurs during the first ≈ 5 Myr. This is to avoid aspects of thermal structure that are inherited from the initial conditions. In the rest of the paper, we therefore refer to the three subsequent subduction stages: (i) *Free sinking* of the slab through the upper mantle: After subduction initiation, the slab sinks rapidly through the upper mantle. The convergence rate at the center of the plate is ≈ 8 cm/yr at the plate center and ≈ 7 cm/yr at the plate edge; (ii) *Interaction with the lower mantle*: As the slab impinges on the more viscous lower mantle (≈ 15 Myr into the model run), it bends backwards on top of the viscosity discontinuity. The convergence rate decreases rapidly, until it reaches ≈ 5 cm/yr at the plate center and ≈ 4 cm/yr at the edge; (iii) *Mature subduction*: Subduction slows down as the slab continues to flatten on the upper-lower mantle boundary. The convergence rate reaches a minimum of ≈ 2 cm/yr at the plate center and ≈ 1 cm/yr at the edge.

This reference model is characterized by slab rollback (20% to 70% of the convergence rate depending on the subduction phase) and, as described, a higher convergence rate at the plate center than the edge (Figure 3a,b). Roll-back induced toroidal flow around the slab edges, from regions of relatively high to relatively low pressure, also enables the trench to deform along-strike (e.g., Funicello et al., 2003; Schellart, 2004; Piromallo et al., 2006). After initiating at an equivalent depth, the DD becomes deeper at the plate edge than at the center, by 5 km after ≈ 50 Myrs (Figure 3c,d).

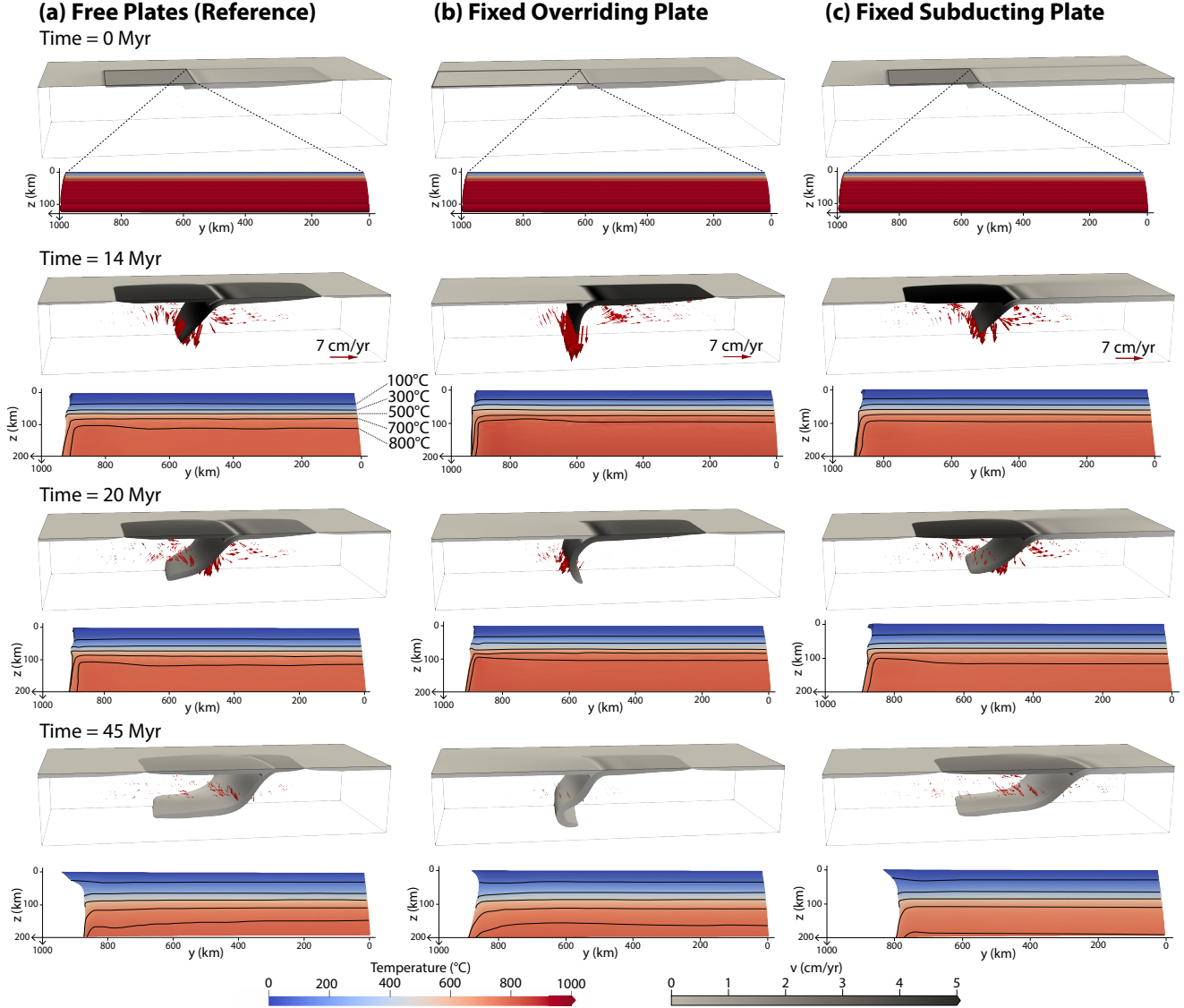


Figure 2: Temporal evolution of models with variable plate mobilities, each with a 2000 km plate width (i.e., “wide” plate models). For each timestep, we show the slab shape, upper mantle velocity field, and the slab-top temperature distribution to a depth of 200 km (with the exception of $t = 0$ Myr, where the slab-top reaches a depth of only 120 km). The black lines show isotherms at 100, 300, 500, 700, 800°C. (a) corresponds to the reference model depicted in Figure 1; (b) corresponds to the wide, fixed overriding plate model; (c) corresponds to the wide, fixed subducting plate model. The grey outline in the top row shows initial the position of the overriding plate.

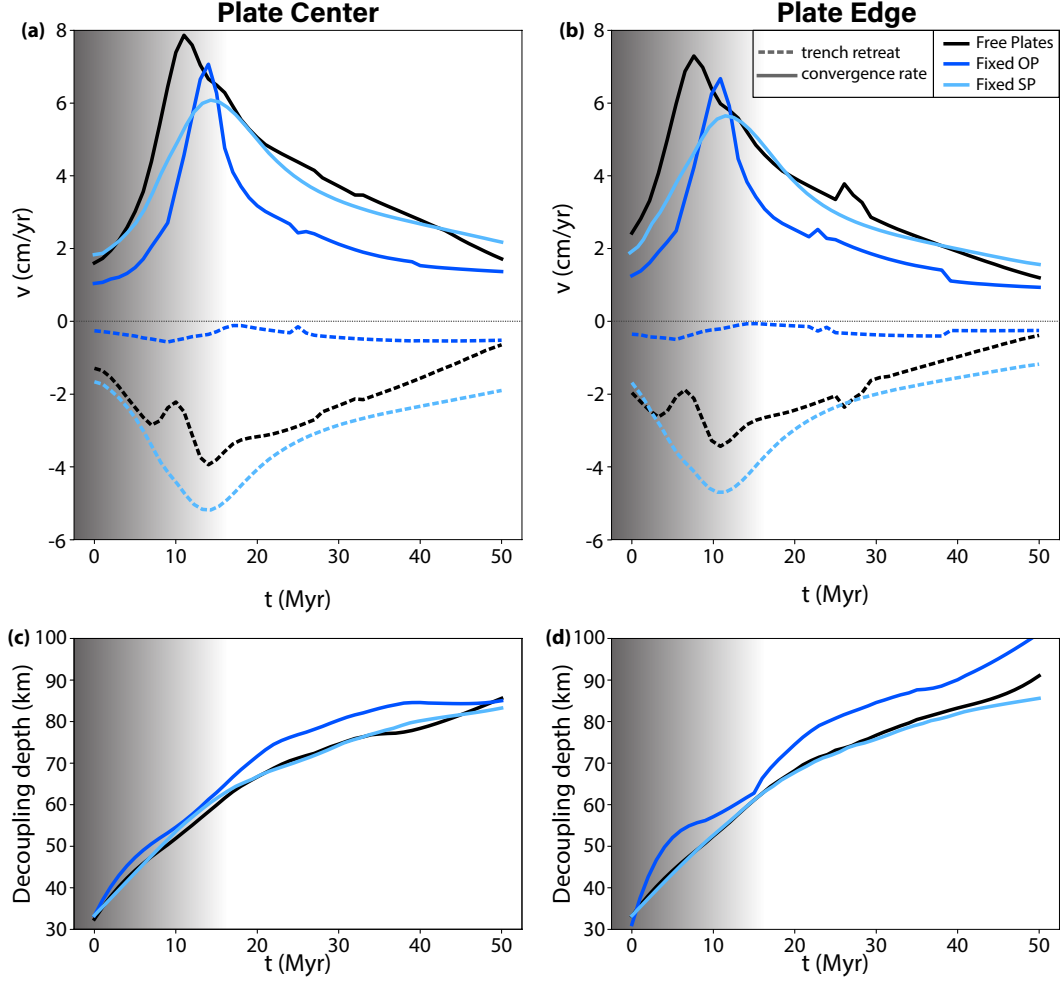


Figure 3: (a, b) Surface kinematics and (c, d) decoupling depth evolution for models with variable plate mobilities, each with a 2000 km plate width. Velocities and decoupling depths are extracted from both the center and edge of the model subduction zones (see Figure 1 for locations). Solid lines and dashed lines correspond to convergence and trench migration rates (negative = trench retreat), respectively. The black lines represents the free plates (reference) case and the shades of blue from dark to light represent, in order, the fixed overriding plate and fixed subducting plate cases. The shaded area indicates the subduction initiation and free sinking phases within the free plates reference model.

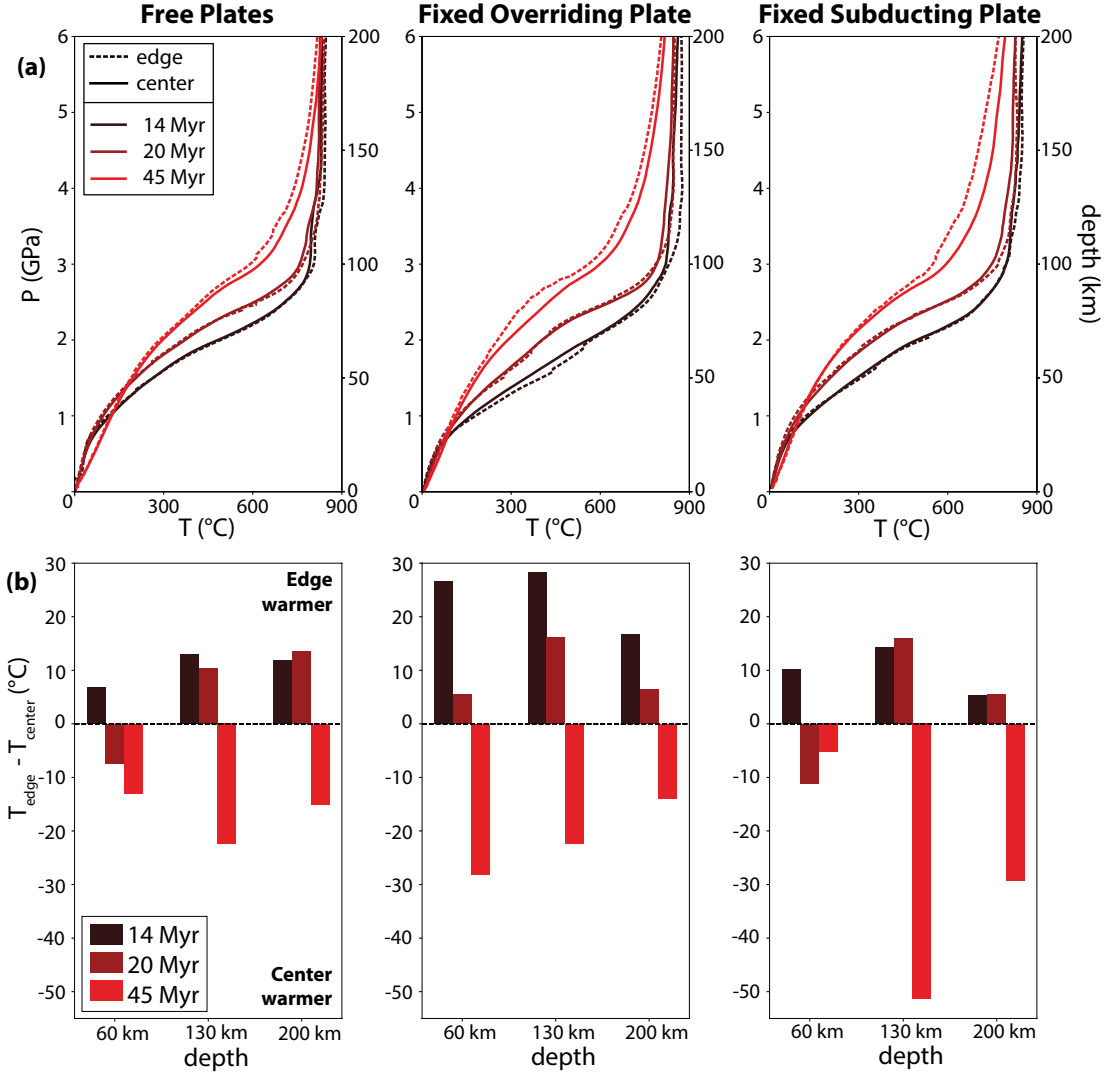


Figure 4: (a) Slab-top pressure-temperature (P-T) conditions for models with variable plate mobilities, each with a 2000 km plate width, and (b) the associated slab-top temperature difference between the slab at the edge and center of the subduction zone at 60, 130, and 200 km depths. In panel (a), dashed lines denote the plate edge P-T conditions and the solid lines the plate center conditions. In panel (b), the different shades of red indicate the three distinct times.

We extract slab-top P-T conditions at the plate center and plate edge, for times within the three characteristic subduction phases, to probe along-strike temperature variation. To avoid the highly deformed area at the very edge of the plate, we extract edge temperatures at a fixed lateral position corresponding to 50 km (in the along-strike direction) from the location of the plate edge at 14 Myr. Despite minor amounts of trench-parallel plate/slab shortening, this distance enables us to always be extract temperatures from within the slab and not the adjacent, flowing mantle. The only exception is the 45 Myr profile for the fixed subducting plate case; intense trench-parallel deformation requires that we extract edge temperatures at ~ 50 km from plate edge at this specific time, in order to have comparable results to the other models/times. The P-T conditions are extracted along the upper surface of the compositional crust (i.e., the upper layer of the slab). As time progresses, the slab-top cools for depths $\ll 200$ km (Figure 4a, Supplementary Figure S2). At depths of ≥ 200 km, cooling is negligible and the near-constant temperature is $\approx 850^\circ\text{C}$. At a given time, slab temperature increases rapidly down to a depth of 50–100 km. At this depth, a kink, most visible for earlier times and located between pressure values of 2–3 GPa (Figure 4), is present in the P-T conditions, which corresponds to a rapid increase in temperature centered on the time-evolving DD (cf. Wada & Wang, 2009; Syracuse et al., 2010).

Slab-top thermal conditions exhibit along-strike variability. To illustrate this, we compute the difference in temperature between plate center and edge for each model, at the same times and three different depths (Figure 4b): 60 km, corresponding to the temperature at or above the DD (depending on the evolving depth of the DD at the particular time), and 130 km and 200 km, corresponding to two depths greater than the DD in which we have viscous coupling between the slab-top and mantle wedge. The absolute temperature difference varies between 5°C and 25°C . In general, the slab edge is warmer than the center of the slab during the first two phases of subduction by $\approx 10^\circ\text{C}$. During the final, mature phase of subduction, this trend reverses as the slab center becomes warmer than the slab edge by $10 - 25^\circ\text{C}$.

3.2 Variable plate mobility

Our reference case consists of two plates that are free to move horizontally (i.e., are not pinned to the model sidewalls). Fixing one of the plates strongly influences trench/slab mobility which, in turn, exerts a significant control on slab evolution and the associated

mantle flow (cf. Christensen, 1996; Stegman et al., 2010). Here, we assess the impact of this variability on along-strike temperature variations.

We first describe the slab dynamics. Regardless of plate mobility, these models have similarities with the free plates reference case: Convergence rate is again lower at the plate edge than at the center (Figure 3a-b) and undergoes a similar pattern of evolution through the three subduction phases. The DD is again deeper at the plate edge than the center (Figures 3c-d) and the DD difference, between the edge and center, increases as the model evolves.

There are, however, key differences with respect to the free plates reference. Most notably trench retreat (Figure 3a-b): In the fixed overriding plate case, trench retreat is negligible (< 1 cm/yr at all times) and, as a result, so is along-strike trench deformation. In the fixed subducting plate case, trench retreat is more rapid than in the reference case throughout the whole subduction process, typically by 20-50%, which enhances trench curvature. Convergence rate also varies with plate mobility (Figure 3a-b): In the fixed overriding plate case, it is up to 60% lower than in the reference case. In the fixed subducting plate case, the convergence rate is again lower than in the reference case for early subduction, but it with similar values during the mature phases. The DD is, in general, greater in the fixed overriding plate case than the free plates reference, by ≈ 10 km, but the fixed subducting plate case has similar DD values. Similarly, the center-to-edge DD increase is greater in the fixed overriding plate case, reaching ≈ 15 km during mature subduction, but this increase is significantly reduced in the the fixed subducting plate case (to 2-3 km of along-strike variation).

Both of the new cases exhibit similar trends regarding slab-top P-T evolution and its variation along-strike. Figure 4 shows that, as in the reference, the slab-top temperature decreases with time, increases with depth, and exhibits a kink in all cases. There is also the same sense of along-strike variability: The edge of the plate is warmer than the center during the early stages of subduction and colder for mature subduction. While the same trends persist, there are variations in the magnitudes of along-strike slab-top temperature changes. In the fixed overriding plate case, the absolute temperature difference varies between 5°C and 30°C , while for the fixed subducting plate case it varies between 5°C and 50°C . This is relative to less variability in the free plates reference ($5-25^{\circ}\text{C}$).

3.3 Variable plate width

We now vary the plate/slab width for both the free plates and fixed overriding plate cases, to examine the effects of three-dimensionality in more detail. We reduce the plate width to examine a narrower subduction zone (1000 km plate width) relative to the reference (2000 km width).

For both cases, the first-order evolution is broadly consistent. As for the wider plate cases, convergence rate remains higher at the plate center than at the edge (Figure 5a-d) and the DD is greater at the plate edge than the plate center (Figures 5e,f). Figure 6 compiles the DD variation in all models with variable plate widths (i.e., the free plates and fixed overriding plate cases); the plate center always has the shallower DD. Vigorous toroidal flow occurs in both free plate cases (Supplementary Figure S3), as both slabs exhibit significant rollback, but is negligible in both fixed overriding plate cases (Supplementary Figure S4). The two clearest differences relate to the convergence rate and DD: Convergence rate is lower in the narrower plate models, the DD is generally deeper, and the magnitude of the along-strike (center-to-edge) DD increase is reduced.

In both narrow subduction zone cases, the P-T conditions again exhibit a decrease in temperature with time (Figure 7a) and along-strike temperature differences (Figure 7b), both following the same trend as the reference model. However, at 20 Myr, the narrow, free plates model exhibits negligible temperature difference along-strike, while for the other two times, the absolute temperature difference is greater than within the reference model (a maximum of $\approx 32^\circ\text{C}$ relative to $\approx 25^\circ\text{C}$ in the reference). The narrow, fixed overriding plate case exhibits a greater temperature difference at 20 Myr ($\approx 30^\circ\text{C}$ maximum) with respect to the equivalent wide plate case ($\approx 15^\circ\text{C}$ maximum), while during the earliest phase, at 14 Myr, this trend is reversed.

3.4 Comparison with two-dimensional cases

To extend the plate width exploration, we constructed equivalent 2-D models for the fixed and free overriding plate cases. 2-D subduction models are equivalent to infinitely long trenches; these tests thus enable us to further isolate the effect of the third dimension. In the free plates model, the 2-D convergence rate is higher ($\approx 10\%$) than that extracted from the center of the 3-D case until the slab starts interacting with the lower mantle; after which, it becomes significantly lower (Supplementary Figure S5a,c).

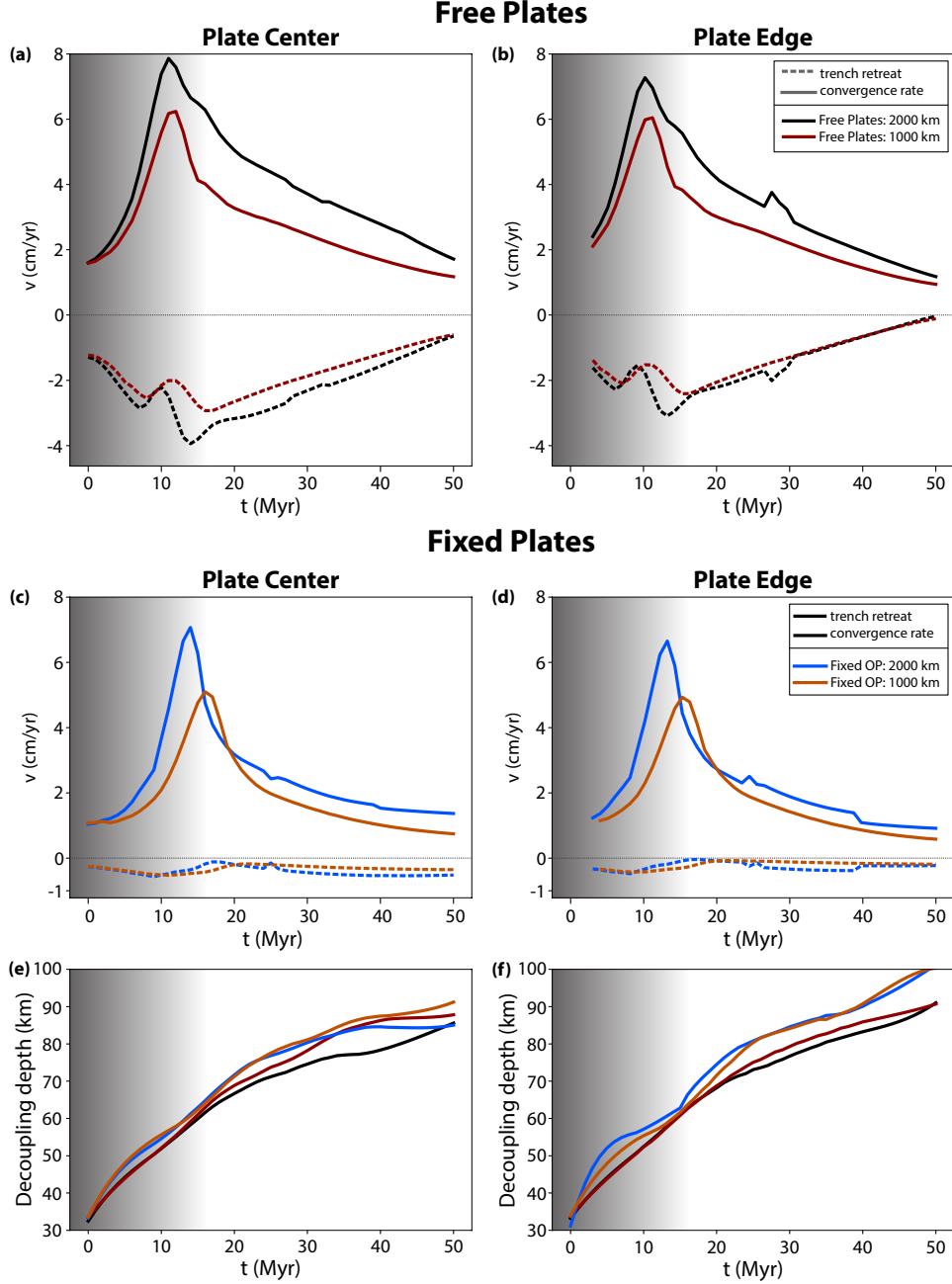


Figure 5: (a – d) Surface kinematics and (e, f) decoupling depth variation for models with variable plate widths, with properties extracted from both the plate center and the plate edge. In panels a-d, solid lines correspond to convergence rate and dashed to trench motion (negative = trench retreat). Dark and light brown lines correspond to correspond to 1000 km wide free plates and fixed overriding plate cases, respectively. Black and blue lines correspond to correspond to 2000 km wide free plates and fixed overriding plate cases, respectively. The shaded area corresponds to subduction initiation and free sinking for free overriding plates. This phase lasts longer for the fixed overriding plate cases.

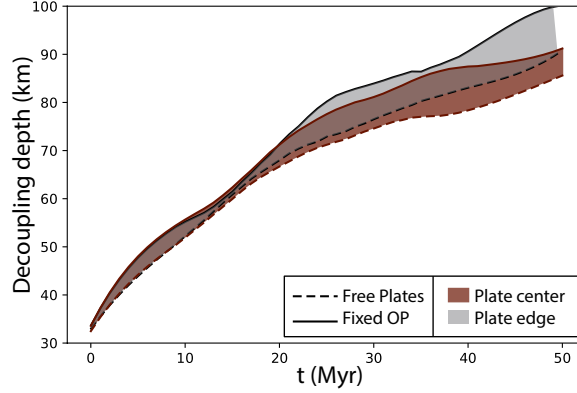


Figure 6: Range of decoupling depths (DD) for all 3-D models with variable plate widths (i.e., free plates and fixed overriding plates cases for 1000 km and 2000 km wide plates). The dashed line corresponds to a free overriding plate model (2000 km wide plates case), which has the minimum DDs, and the solid line to a fixed overriding plate model (1000 km wide plate case), which has the maximum DDs. The range of plate edge DDs is in grey and the range of plate center DDs are in brown. Slabs at the plate edge always have a greater DD than at the center.

The DD is deeper than within the equivalent 3-D model (both at the center and edge of the 3-D slab). In the fixed overriding plate case, both the convergence rate and DD are similar to the 3-D model (Figure S5b,d).

We compute the temperature difference between slab-tops in the 2-D and 3-D cases; in the 3-D case, we use the slab-top temperature extracted from the plate center (Figure 8). The overall trend is independent of overriding plate mobility: for early stages and shallow depths (60 km), the slab-top is warmer in the 3-D models by 9–25°C. For all other times and depths, the 2-D slab-top is warmer by 7–35°C.

4 Discussion

Our models highlight an along-strike gradient in slab-top temperature that evolves through time. We start by addressing our modeled subduction zone evolution – particularly the convergence rate, decoupling depth (DD), and mantle flow field – and show how these quantities combine to dictate the slab-top thermal evolution. We then compare our modeling results to previous modeling studies and natural observations.

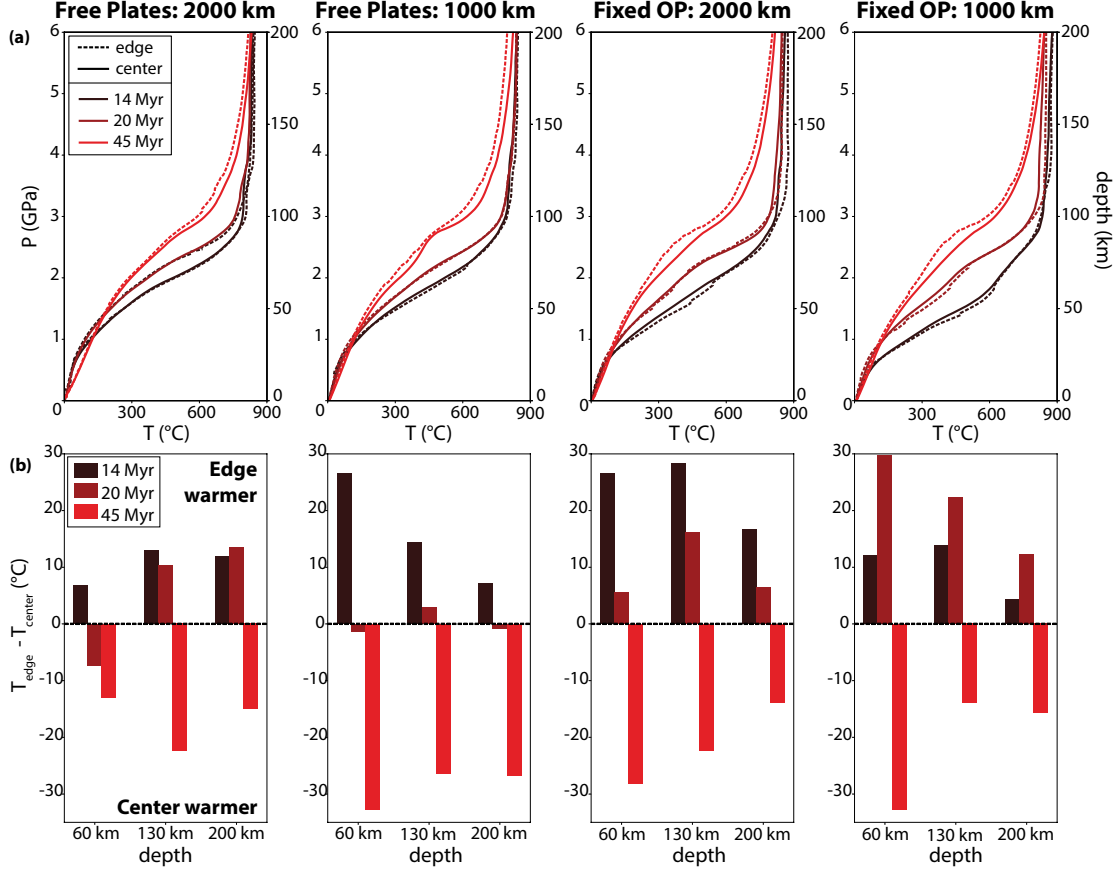


Figure 7: (a) Slab-top P-T conditions for models with variable plate widths, and (b) the associated temperature difference between the slab-top at the edge and center of the subduction zone at 60, 130, and 200 km depths. In panel (a), dashed lines denote the plate edge P-T conditions and the solid lines the plate center paths. In panel (b), different shades of red indicate the three distinct times.

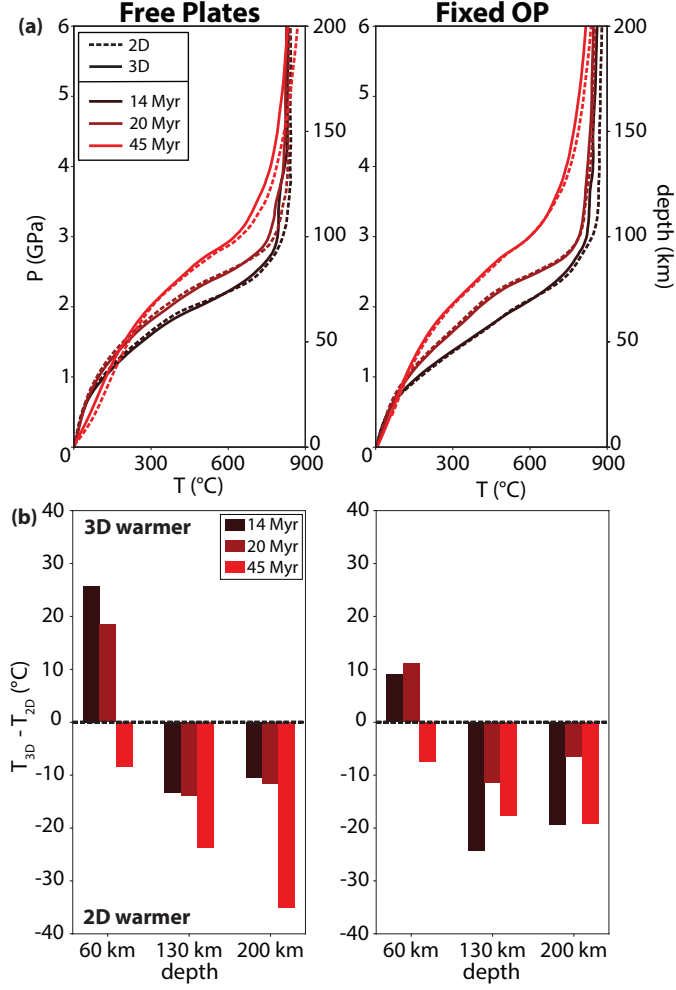


Figure 8: (a) P-T conditions for the 2-D vs 3-D cases, and (b) slab-top temperature differences between 2-D and 3-D models at 60, 130, and 200 km depths. In a), the solid line denotes the 3-D case; the dashed lines the 2-D case. The different shades of red indicate the three distinct times.

4.1 Slab dynamics

The temporal evolution of our reference model, with free and 2000 km-wide plates, is characterized by slab rollback, vigorous toroidal flow at depth, and along-strike variability in trench retreat and convergence rates. The convergence rate is greater at the plate center than the plate edge, by $\approx 10\%$, which strongly affects the DD. As the model progresses, the DD tends to deepen due to the thermal aging of the overriding plate and the associated development of a cold forearc wedge corner. On the other hand, a higher convergence rate increases the vigor of poloidal flow generated by slab sinking (Sleep & Toksöz, 1971), which, in turn, produces more efficient transport of hot mantle material into the wedge corner. Since the DD marks the transition from the cold forearc nose to the warm mantle wedge, this vigorous transport pushes the DD towards the surface (cf. Arcay et al., 2007), counteracting the effect of thermal cooling to some extent. At the plate center, where convergence rates are greater, the DD is therefore shallower than at the plate edge in all of the models. The edge-to-center DD difference increases with time, and reaches up to 15 km at the end of the run (Figure 6). As this edge-to-center DD difference increases with time, so does the average DD, from 30 km at the start to ≈ 90 km at the end of the model (cf. Holt & Condit, 2021). Surface heat flux measurements suggest an average DD of 60 – 80 km at present-day subduction zones (Furukawa, 1993; Wada & Wang, 2009), and recent geodetic measurements in Honshu, Japan suggest that the DD can indeed vary along-strike by ≈ 25 km (Dhar et al., 2022). This gives us confidence in the natural applicability of the first-order features of our DD evolution and the associated thermal effects (Sections 3.1-3.3). We note that we set the DD limit, or “maximum decoupling depth (MDD)”, by the 100 km depth at which we eliminate our weak crust and thereby couple the slab to mantle wedge. In reality, the MDD is likely dictated by a switch from rheologically weak hydrated to stronger anhydrous rocks in either the crust or mantle wedge (Hirauchi & Katayama, 2013; Agard et al., 2020; Peacock & Wang, 2021; Kerswell et al., 2021).

Fixing the overriding plate reduces slab rollback/trench retreat (e.g., Capitanio et al., 2010; Schellart & Moresi, 2013; Holt et al., 2015). This results in almost no along-strike trench deformation, negligible toroidal flow, and greater dips by up to 15° . Conversely, fixing the subducting plate promotes elevated slab rollback, more vigorous toroidal flow, and reduces dips by $\approx 10^\circ$. Relative to the reference, convergence rate is lower in the fixed overriding plate case, by up to 2 cm/yr, which results in a greater DD. Impor-

tantly, in all models, the convergence rate is higher at the plate center than at the edge; this produces the observed DD increase from plate center to edge.

Varying the plate width does not modify the first-order slab dynamics or the relationship between convergence rate and DD. Reducing the plate width lowers the average convergence rate (Stegman et al., 2010) and, in turn, deepens the DD. The most significant effect of increasing the effective plate width (i.e., 2-D models/infinite plate) occurs during mature subduction (i.e., after slab interaction with 660 km) in the free plates model. During this phase, the 2-D convergence rate is reduced relative to the equivalent 3-D case; as expected, this results in an increased DD.

4.2 Spatio-temporal Thermal Variability

In our models, spatio-temporal variability in convergence rate is the main control on along-strike slab-top temperature variation. The product of convergence rate and plate age (and sometimes the sine of slab dip) is termed the “thermal parameter” (Kirby et al., 1996) and often used as an approximate indicator of slab-top temperatures in models (e.g., van Keken et al., 2011) and natural subduction zones (e.g., Wiens, 2001). High values, e.g. due to rapid convergence rates, correspond to cold slabs. Maunder et al. (2019) explored how well the thermal parameter predicts slab temperatures within models with different decoupling depths (DD). Because the DD dictates the down-dip extent of cold material that blankets the slab, they found that a deeper DD exerts a strong slab cooling effect (cf. Wada et al., 2008) and hence proposed the DD should be incorporated into updated formulations of the thermal parameter. This new description is well suited to characterizing the slab-top temperatures in our 3-D models which, as described in previous sections, exhibit significant spatio-temporal variability in not only convergence rate but also the DD. In our dynamic models, the DD is controlled by both the gradual cooling of the overriding plate (i.e., a gradual deepening of the DD) and variability in convergence rate (i.e., a greater rate produces more vigorous mantle flow that resists this deepening of the DD).

In addition to controlling temperature via the rate of slab sinking - as expressed by the classical thermal parameter - this link between convergence rate and DD results in additional along-strike variability in slab-top temperature. These two mechanisms dom-

inate at different stages of subduction and produce the following evolution in along-strike slab-top temperature variation (Figure 9).

Early stages During the initial phases of subduction (up to 20 Myr), the convergence rate is high (up to 8 cm/yr during free sinking) and reaches the maximum value at the slab center, regardless of plate width and mobility. Following the classical thermal parameter, a higher convergence rate at the slab center corresponds to rapid advection of the cold slab (relative to warming via thermal diffusion) and so reduced temperatures. Hence, the slab is cooler at the plate center than at the plate edge ($T_{\text{center}} < T_{\text{edge}}$) for all models and at all depths.

Mature subduction After the slab interacts with the lower mantle (at ≈ 20 Myr), the average convergence rate decreases by up to 70%, but remains higher at the plate center than at the edge. Because convergence rates are overall much lower, the basic slab sinking effect, as described for the early stages, is overridden by the effect of the DD. At this stage, enough time has elapsed for the DD to be highly variable along-strike (by 5–15 km, depending on the model). This is because, at the plate edge, a lower convergence rate results in less vigorous poloidal flow and hence a greater DD relative to the plate center (where more vigorous flow resists deepening of the DD). Thus, as the DD increases at the plate edge, bringing colder isotherms to greater depths, the temperature in this region decreases. The along-strike temperature trend, therefore, reverses: The plate center becomes warmer than the plate age ($T_{\text{center}} > T_{\text{edge}}$). This effect is influential at all of the analyzed depths but is generally most pronounced at shallower depths near the DD (i.e., 60 km relative to 130 km and 200 km).

The vigor of toroidal flow has been proposed to exert a strong effect on slab temperatures near lateral slab edges. Modeling studies predict toroidal flow concentrated around the slab edges (e.g., Piromallo et al., 2006; Jadamec & Billen, 2010; Faccenda & Capitanio, 2012) and that the associated influx of hot mantle material increases the slab-top temperature (e.g., Kincaid & Griffiths, 2003; Kneller & van Keken, 2008; Plunder et al., 2018). However, in our models, the presence of toroidal flow has a limited effect on slab-top temperature variation: when the overriding plate is fixed and toroidal flow is almost absent (Supplementary Figure S6), the general sense and magnitudes of temperature variation are comparable to the free plates reference with substantial toroidal flow. Only when the subducting plate is fixed – and subduction is rollback dominated

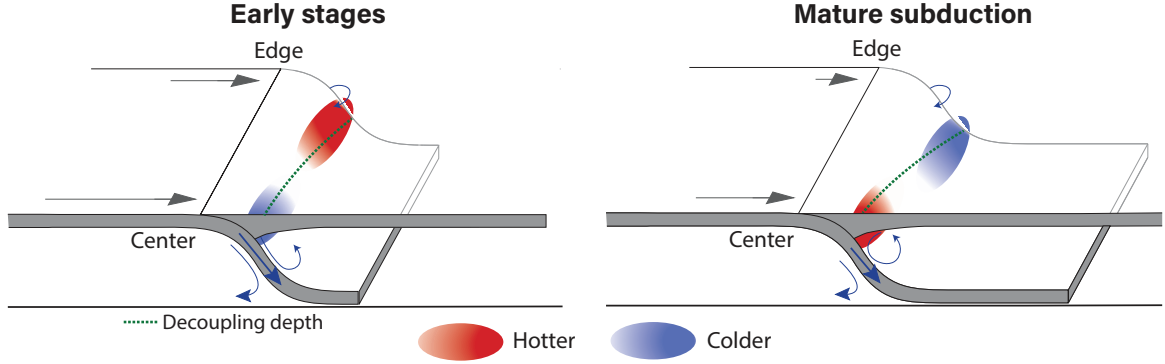


Figure 9: Illustration of the main study results. The length of the dark gray arrows indicates the magnitude of the plate velocity and the green dashed lines indicated the position of the decoupling depth.

and toroidal flow is especially vigorous (Supplementary Figure S6) – is there a clear signature of toroidal flow-induced heating. During the intermediate phase (20 Myr), the along-strike temperature change is $\approx 30\text{--}50\text{ }^{\circ}\text{C}$ at great depths (130, 200 km); this is elevated relative to the $\approx 15\text{--}25\text{ }^{\circ}\text{C}$ temperature difference in the other two cases with free subducting plates (Figure 4).

4.3 Comparison with previous modeling studies

In our dynamic models, subduction properties evolve self-consistently with time. Properties that are key with respect to the thermal field include the interlinked velocity field, decoupling depth, and slab shape; their dynamic variability results in a highly time-dependent thermal evolution. Because of this dynamism, it is challenging to compare our results with the kinematically-driven and fixed slab geometry models that are typically used to investigate slab thermal structure. A first-order comparison, however, is useful to place our models in the context of previous work.

Previous kinematically-driven modeling studies have shown that a greater DD results in lower slab-top temperatures (Syracuse et al., 2010; Perrin et al., 2018; Maunder et al., 2019, Wang et al., 2023). In addition, previous time-dependent 2-D models exhibit a gradual deepening of the DD, due to the conductive cooling of the overriding plate (with mantle wedge flow resisting this DD increase to some extent). Taken together,

this causes a gradual cooling of the slab-top from early to mature subduction (Kincaid & Sacks, 1997; Holt & Condit, 2021; Zhou & Wada, 2021). Our 3-D models exhibit the same trend but enable us to probe the time dependency of slab thermal structure in the third (along-strike) dimension. Our key finding is that the plate center is cooler than the plate edge for early subduction and warmer for mature subduction. Before we compare our modeled along-strike thermal variability with existing 3-D modeling studies, we note that the slab-top temperature difference between our 3-D models (at the plate center) and equivalent 2-D models is broadly equivalent to the benchmark models of Bengtson & van Keken (2012). Below the DD, slab-tops in our 2-D models are up to 35°C warmer than the equivalent 3-D case. In Bengtson & van Keken (2012), the equivalent values are $\approx 50^\circ\text{C}$, with the discrepancy likely due to a more curved trench in this previous study.

Although our models highlight a new behavior with regards to slab-top temperature variability - i.e., a flip in the sense of along-strike thermal variation - the along-strike variation in slab-top temperature ($\leq 50^\circ\text{C}$) is on the low end of the range commonly cited in the modeling literature (from 50°C to 200°C; e.g., Bengtson & van Keken, 2012, Morishige & van Keken, 2014, Ji et al., 2016, Plunder et al., 2018). In these studies, toroidal flow is invoked as a primary driver of along-strike temperature variation but, in our models, toroidal flow only has a clearly identifiable effect on slab-top temperature at great depths ($\geq 130\text{ km}$) within the rollback dominated, fixed subducting plate case (Figures 2c, 4, S6). We suspect that this difference is due to convergence obliquity relative to trench strike (e.g., Wada et al., 2015; Ji et al., 2016; Plunder et al., 2018) and/or high trench curvature (e.g., Bengtson & van Keken, 2012) in most of these previous models. Higher obliquity and curvature both drive more vigorous trench parallel flow and hence result in greater along-strike temperature differences. In complex tectonic situations, with highly nonlinear slab shapes and/or multiple nearby slabs, toroidal or trench parallel flow can be particularly vigorous and hence produce even greater along-strike temperature anomalies (e.g., Araya Vargas et al., 2021; Wada et al., 2015; Ji et al., 2017; Wada & He, 2017). Additional tests in some of these studies (e.g., Bengtson & van Keken, 2012; Plunder et al., 2018), however, consider subduction zones with lower trench curvature and subduction obliquity; this results in less vigorous trench parallel flow and hence smaller temperature variations (e.g., $\approx 30\text{--}50^\circ\text{C}$), as is consistent with the $\leq 50^\circ\text{C}$ of along-strike slab variability in our models. We also note that these pre-

vious models do generally exhibit the same along-strike trend as our models during the mature subduction phase (i.e., the slab-top is warmer at the center than the edge of the subduction zone).

The analog models of Kincaid & Griffiths (2003, 2004) are often also used as evidence for the pronounced effect of toroidal flow on slab temperatures. They are most comparable to our mature subduction stage and predict along-strike temperature differences of 35°C to 120°C and, as in our models during this phase, cooler slab edges if trench retreat is dominant. In these models, a temperature increase at the center of the slab is attributed to the focusing of toroidal return flow around the slab and into the slab center. Again, the magnitude of along-strike temperature variation is of the same sign, but generally of greater magnitude than our modeled along-strike temperature variation during mature subduction, which we primarily attribute to along strike DD variations as opposed to toroidal flow. In this case, the discrepancy is not due to oblique subduction or curved trenches as Kincaid & Griffiths (2003, 2004)’s models, like ours, have straight trenches and near-orthogonal convergence. Rather, toroidal flow is again more vigorous in Kincaid & Griffiths models; we suspect this is partially due to their slabs being rigid and hence unable relieve a component of dynamic pressure build-up via slab deformation (cf. Kincaid & Olson, 1987). Subduction is also entirely rollback driven and the plates are narrower (650 km width), relative to our 1000-2000 km wide plates and 20 – 70% of convergence accommodated by rollback in our models with free subducting plates. In our models, with wider plates and generally less rollback, less toroidal flow (and hence hot mantle material) reaches the plate center (Supplementary Figure S7), thereby driving less slab-top heating due to this mechanism. However, despite this discrepancy in the magnitude of thermal variability, Kincaid & Griffiths (2003, 2004) also highlight a correlation between lower subduction velocities and a greater DD, and hence cooler slab-top temperatures, as found to be important within our models.

To summarize, our models suggest a relatively limited role for toroidal flow-induced slab heating - and hence reduced along-strike slab-top temperature variation - in tectonic situations where trenches are relatively wide, straight, and accommodate near-orthogonal convergence, and where slabs are able to deform. However, when convergence rates are free to evolve in time and along-strike, the sense of variation may be similar to previous modeling studies during mature subduction (i.e., a warmer slab center), albeit driven by a distinct mechanism (i.e., along-strike variation in the DD).

4.4 Comparison with natural cases

The modeled along-strike changes in slab-top temperature are reflected, to first order, in geological observables extracted from some Earth subduction zones. Here, we provide a concise overview of a few such observations, but emphasize that our models correspond to highly idealized subduction. Natural comparisons are therefore mainly limited to qualitative trends.

The gradual cooling of the slab-top in our time-dependent models is supported by the exhumed rock record. Exhumed high pressure/low temperature oceanic rocks highlight extremely variable P-T-t conditions with, in general, warmer conditions during the early stages of subduction and an overall cooling trend as subduction progresses to more mature phases (e.g. Platt, 1975; Cloos, 1985; Krebs et al., 2008; Krebs et al., 2011; Agard et al., 2018; Dragovic et al., 2020; Takeshita et al., 2023). This is consistent with the evolution of slab-top P-T conditions in our models (Supplementary figure S8). These rocks reach peak pressures of up to 2.5 GPa (e.g. Beaumont et al., 1999; Jolivet et al., 2003; Yamato et al., 2007; Agard et al., 2009), and therefore correspond to depths of up to 100 km; at such depths in our models, the slab-top indeed cools rapidly.

Our models predict along-strike variability in slab-top temperatures, with a trend that varies from early to mature subduction zone. Although the modeled temperature contrast is low ($\leq 50^\circ\text{C}$), the sense of variation agrees with that inferred from along-strike variations in exhumed rocks and arc magma geochemistry at some localities. During the earlier phases, our models predict colder slab centers due to elevated plate convergence. This is compatible with exhumed rock P-T constraints in the Aegean (in the Miocene) and Franciscan (in the late Jurassic-early Cretaceous) where, respectively, peak temperatures decrease by 100°C (Jolivet et al., 2010) and up to 300°C (e.g., Wakabayashi & Dumitru, 2007; Ernst, 2011) towards the center of the system. Along-arc variation in magma geochemistry in the Sangihe arc has also been linked to a warmer slab edge (Hanyu et al., 2012). During mature subduction, where our models predict a warmer slab center due to shallower DD, exhumation is typically linked to transient processes, such as large-scale changes in tectonic activity or slab breakoff (e.g., Wang et al., 2023; Agard et al., 2018); isolating the modeled signal, even if present, is therefore challenging. However, the modeled, mature thermal variability is broadly consistent with that inferred from arc magma geochemistry in the Central Kamchatka Depression, with a plate cen-

ter proposed to be $\approx 150^\circ\text{C}$ warmer than the edges (Portnyagin & Manea, 2008). We also inspected whether our modeled along-strike variations in DD would have a measurable effect on surface flux values in the forearc region of the overriding plate. There is negligible along-strike heat flux variation during the early subduction stages, when DD values are similar at the plate edge and center (Supplementary Figure S9). However, during mature subduction, forearc heat flow at the edge of the subduction zone is $\sim 10 \text{ mW/m}^2$ lower than that at the plate center, as a result of the $\sim 5 \text{ km}$ greater DD at the plate edge. This relationship between surface heat flow and DD agrees with Wada & Wang (2009) but, given measurement scatter and other sources of heat flow variability, the along-strike variation is likely too small to effectively use surface heat flow to detect DD variations of the magnitudes produced in our models.

Returning to along-strike trends in slab-top temperatures, there are, however, multiple subduction zones that do not match these trends; e.g., the Hikurangi Trench where Yabe et al. (2014) suggest that slab-top temperatures vary uniformly from edge to edge. The absolute values of modeled temperature variations are also much lower than these observational estimates. This may be due to the occurrence of more complex subduction geometries than within our models (and so more slab heating associated with toroidal flow) and/or the greater along-strike convergence rate changes (and so enhanced versions of the effects modeled here). Thus, further tailoring slab geometries and plate convergence to natural cases - and also incorporating additional mechanisms of heat transport and production like small-scale convection (e.g., Honda & Saito, 2003, Davies et al., 2016) and shear heating (e.g., Molnar & England, 1990, Gao & Wang, 2014) - would aid with a more direct comparison to specific cases. Shear heating, in particular, would also weaken the plate interface which could, in turn, elevate convergence rates (e.g., Behr & Becker, 2018) and therefore affect slab-top temperature distribution. Our models, however, present an initial, idealized framework for the spatio-temporal variability of slab-top temperature.

5 Conclusions

Geological observations suggest that slab-top thermal structure simultaneously varies along strike and through time. To develop intuition about the drivers of this thermal variability, we developed time-dependent and dynamic 3-D subduction models. Specifically, we investigated the time dependence of the magnitude and sign of along-strike temper-

ature variation for models with variable plate/slab widths (1000 km, 2000 km, infinite) and plate mobilities (fixed vs. free subducting and overriding plates).

The along-strike temperature variation is heavily time-dependent but consistent between our models. In all cases, the region of the slab at the center of the subduction system is colder than the slab edge by $\leq 35^\circ\text{C}$ during the early stages of subduction stages. During mature subduction, this trend reverses; the slab center becomes warmer than the slab edge by $\leq 50^\circ\text{C}$. During the early stages, the colder slab center is due to more rapid convergence rates - and so less diffusive heating of the slab - at the center of the subduction zone. During the mature stages, the slab center is warmer as higher convergence rates begin to push the decoupling depth (DD), and hence warm mantle material, to shallower depths than at the edge. Relative to some kinematically-driven subduction modeling studies, our dynamic models generally predict lower along-strike temperature changes (tens instead of hundreds of degrees). This is due to a reduced role for toroidal flow-induced heating of the slab center; deforming slabs, relatively wide trenches, and near-orthogonal convergence reduce this effect in our models.

Relative to temperatures reconstructed from geological observations, our models generally predict comparable trends but reduced magnitudes of variation. More vigorous toroidal flow (e.g., due to more complex slab geometries) or greater along-strike variability in convergence rate (i.e., to produce more extreme DD variations) would likely reconcile this. Nevertheless, our models highlight new controls on 3-D slab-top temperatures that can only emerge with dynamic models: an important role for along-strike DD variations, and a time-dependent evolution of the sense of along-strike temperature variation.

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Data Availability Statement

The ASPECT plugin, input geometries and parameter files necessary to run the models can be found in the following Zenodo repository: <https://doi.org/10.5281/zenodo.8075606>

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